

Influence of Temperature on the Interface Chemistry of $\text{Li}_x\text{Mn}_2\text{O}_4$ Electrodes

T. Eriksson,^{†,§} A. M. Andersson,[†] C. Gejke,[‡] T. Gustafsson,[†] and J. O. Thomas^{*,†}

Department of Materials Chemistry, Ångström Laboratory, Uppsala University, Box 538, SE-751 21 Uppsala, Sweden, and Department of Applied Physics, Chalmers University of Technology, SE-412 96 Göteborg, Sweden

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The temperature dependence of surface layer formation on $\text{Li}_x\text{Mn}_2\text{O}_4$ electrodes in carbonate-based electrolytes has been studied. No significant differences were observed in the elemental composition of the surface film for cycled and stored samples. This argues against an electrochemical contribution to the surface film formation at elevated temperature. A surface film is formed at higher temperatures containing poly(oxyethylene)/polycarbonate, LiF, $\text{Li}_x\text{PO}_y\text{F}_z$, and phosphorus oxides (or $\text{Li}_x\text{BO}_y\text{F}_z$ and boron oxides, depending on the electrolyte salt). The thickness and coverage increase at higher temperatures. No onset temperature could be found for the formation process, suggesting a general increase in reaction kinetics with temperature. A model is presented for the surface layer formed on $\text{Li}_x\text{Mn}_2\text{O}_4$ ($0 \leq x \leq 1$) electrodes in contact with carbonate-based electrolytes. Polymeric compounds were found to precipitate closest to the electrode surface, with an intermediate layer of LiF and a phosphorus-rich layer outermost.

Introduction

LiMn_2O_4 has been one of the most exploited electrode materials in Li ion battery applications through its ability to release and reinsert lithium ions from the host lattice reversibly, as found by Thackeray and co-workers in 1983.^{1,2} LiMn_2O_4 has a cubic spinel structure with space group *Fd3m*; Li occupies the 8a-site, Mn the 16d-site, and O the 32e-site. There have since been many synthetic, structural, and electrochemical investigations of the material.^{3–10} Surface film formation has recently been reported on active cathodes; the interface between LiMn_2O_4 cathodes and various electrolytes has received special attention.^{9–15} Electrochemical reactions occur at the

interface, and it is thus of crucial importance to understand the cathode/electrolyte interface to avoid parasitic side reactions on both the electrolyte and cathode sides.

For organic liquid and solid polymer electrolytes, a solid electrolyte interphase (SEI) forms on the electrode surface at low potentials vs Li/Li^+ during the first reduction step; this can even occur spontaneously in the case of alkali-metal electrodes (e.g., Li).^{16,17} This SEI layer has been analyzed in detail,^{18–22} and it is generally agreed that the SEI layer on the graphite or Li metal anode of the Li ion battery comprises a mixture of insoluble Li salts and organic species, with the salt precipitating closest to the electrode surface.²³ This heterogeneous film coating constitutes a physical diffusion barrier for solvent species close to the electrode surface, efficiently preventing further reduction of the electrolyte.

At room temperature, X-ray photoelectron spectroscopy (XPS) and infrared spectroscopy (FTIR) results have consistently given evidence of LiF, Li_2O , Li_2CO_3 , Li-alkyl carbonates, and electrolyte salt reduction products (e.g., Li_xPF_y , $\text{Li}_x\text{PO}_y\text{F}_z$, and POF_3 , if LiPF_6 is used) as components of the SEI film.^{18–23} Other species such as polymers, Li–O–C, and polycarbonates have also been reported.^{24,25} At elevated temperatures, the anode SEI film undergoes changes which are observed as exothermic peaks in

[†] Uppsala University.

[‡] Chalmers University of Technology.

[§] Present address: Materials Sciences Division, Lawrence Berkeley National Laboratory, University of California, Berkeley, CA 94720.

* To whom correspondence should be addressed: e-mail josh.thomas@mkem.uu.se.

(1) Thackeray, M. M.; David, W. I. F.; Bruce, P. G.; Goodenough, J. B. *Mater. Res. Bull.* **1983**, *18*, 461.

(2) Thackeray, M. M.; Johnson, P. J.; de Picciotto, L. A.; Bruce, P. G.; Goodenough, J. B. *Mater. Res. Bull.* **1984**, *19*, 179.

(3) Ohzuku, T.; Kitagawa, M.; Hirai, T. *J. Electrochem. Soc.* **1990**, *137*, 769.

(4) Tarascon, J. M.; Wang, E.; Shokoohi, F. K.; McKinnon, W. R.; Colson, S. *J. Electrochem. Soc.* **1991**, *138*, 2859.

(5) Tarascon, J. M.; McKinnon, W. R.; Cowar, F.; Bowmer, T. N.; Amatucci, G.; Guyomard, D. *J. Electrochem. Soc.* **1994**, *141*, 1421.

(6) Ammundsen, B.; Jones, D. J.; Rozière, J.; Burns, G. R. *Chem. Mater.* **1996**, *8*, 2799.

(7) Thackeray, M. M. *Prog. Solid State Chem.* **1997**, *25*, 1.

(8) Liu, W.; Kowal, K.; Farrington, G. C. *J. Electrochem. Soc.* **1998**, *145*, 459.

(9) Nishizawa, M.; Uchiyama, T.; Itoh, T.; Abe, T.; Uchida, I. *Langmuir* **1999**, *15*, 4949.

(10) Uchiyama, T.; Nishizawa, M.; Itoh, T.; Uchida, I. *J. Electrochem. Soc.* **2000**, *147*, 2057.

(11) Antonini, A.; Bellito, C.; Pasquali, M.; Pistoia, G. *J. Electrochem. Soc.* **1998**, *145*, 2726.

(12) Aurbach, D.; Levi, M. D.; Levi, E.; Teller, H.; Markovsky, B.; Salitra, G.; Heider, U.; Heider, L. *J. Electrochem. Soc.* **1998**, *145*, 3024.

(13) Aurbach, D.; Gamolsky, K.; Markovsky, B.; Salitra, G.; Gofer, Y.; Heider, U.; Oesten, R.; Schmidt, M. *J. Electrochem. Soc.* **2000**, *147*, 1322.

(14) Du Pasquier, A.; Blyr, B.; Courjal, P.; Larcher, D.; Amatucci, G.; Gérard, B.; Tarascon, J.-M. *J. Electrochem. Soc.* **1999**, *146*, 428.

(15) Matsuo, Y.; Kostecki, R.; McLarnon, F. *J. Electrochem. Soc.* **2001**, *148*, A687.

(16) Peled, E. *J. Electrochem. Soc.* **1979**, *126*, 2047.

(17) Peled, E.; Golodnitsky, D.; Ardel, G.; Eshkenazy, V. *Electrochim. Acta* **1995**, *40*, 2197.

(18) Aurbach, D.; Weissman, I.; Schechter, A.; Cohen, H. *Langmuir* **1996**, *12*, 3991.

(19) Schechter, A.; Aurbach, D.; Cohen, H. *Langmuir* **1999**, *15*, 3334.

(20) Aurbach, D.; Markovsky, B.; Schechter, A.; Ein-Eli, Y.; Cohen, H. *J. Electrochem. Soc.* **1996**, *143*, 3809.

(21) Kanamura, K.; Tamura, H.; Takehara, Z. *J. Electroanal. Chem.* **1992**, *333*, 127.

(22) Kanamura, K.; Tamura, H.; Shiraishi, S.; Takehara, Z. *J. Electroanal. Chem.* **1995**, *394*, 49.

(23) Aurbach, D. In *Nonaqueous Electrochemistry*; Aurbach, D., Ed.; Marcel Dekker: Basel, 1999; Chapter 6.

(24) Bar-Tow, D.; Peled, E.; Burstein, L. *J. Electrochem. Soc.* **1999**, *146*, 824.

(25) Genies, S.; Yazami, R.; Garden, J.; Frison, J. C. *Synth. Met.* **1998**, *93*, 77.

differential scanning calorimetry (DSC) studies.²⁶ Melting of the original SEI layer, dissolution of soluble species,²⁷ and transformation of the metastable semicarbonates into stable Li_2CO_3 have all been suggested.²⁸ A transformation of Li_2CO_3 in the anode SEI was recently shown to be responsible for irreversible capacity losses after storage of graphite cells at elevated temperatures.²⁹

We have recently conducted an extensive spectroscopic investigation of the surface film formation on LiMn_2O_4 cathodes during cycling and storage in alkyl carbonate electrolytes at ambient temperature.³⁰ It was found that the surface film consisted of electrolyte solvent and salt decomposition products, LiF , phosphorus oxides (if LiPF_6 was used as electrolyte salt), and poly(oxyethylene). The reactions leading to the formation of these species were found to be related to changes in the surface structure of LiMn_2O_4 and the formation of a Li-rich cathode surface. A mixture of the solvents ethylene carbonate (EC) and dimethyl carbonate (DMC) has been reported to produce the best passivation layer for active electrodes;¹⁹ we have therefore focused on these solvents in our cathode surface film studies.

Another aspect of utmost relevance for the functionality and safety of the electrode/electrolyte system in a Li ion battery is its elevated temperature stability; this is typically a well-known trouble spot in LiMn_2O_4 electrodes.^{31,32} Improvements in the electrochemical properties of LiMn_2O_4 electrodes at ambient and elevated temperatures have resulted from cation doping on the Mn 16d-site in the structural lattice.³³ Doping with Al^{3+} increases the average Mn oxidation state, the voltage, and stabilizes the host lattice by decreasing the cell parameter.³⁴ While this reduces the fraction of $\text{Mn}^{3+}/\text{Mn}^{4+}$ redox couples, thereby lowering the capacity of the material, doping with F on the O 32e-site improves the capacity.³⁴ Efforts have also been made to improve the interface properties of the material,^{35,36} but there would seem to be no consensus as to the interface reactions at the electrode surface at elevated temperatures. To the best of our knowledge, this present study is the first to identify the species formed at a Li ion battery composite cathode interface at elevated temperatures.

Experimental Section

Composite LiMn_2O_4 electrodes were made by extrusion of a slurry containing 80 wt % LiMn_2O_4 (Merck, Selectipur), 15 wt % Shewinigan Black carbon powder (CB), and 5 wt % EPDM (ethylene propylene diene terpolymer) binder dissolved in cyclohexane onto an Al foil. The EPDM binder was used, instead of the more widely used PVdF binder, to avoid overlap with F peaks in the XPS spectra.

All electrochemical cycling was performed using flat, flexible, laminated cells. The electrolyte was 1 M salt in EC/DMC 2:1 v/v; two different salts were used: LiPF_6 (Selectipur, Merck) and

LiBF_4 (battery grade, Tomiyama). See ref 29 for more information on preparation and assembly procedures.

The cells were cycled at a C/3 rate, corresponding to a current density of about 0.15 mA/cm². The potential cutoff was 3.5 and 4.3 V vs Li/Li^+ . The electrodes from the cycled cells were compared to a series of electrodes extracted from stored cells made in exactly the same way as the cycled cells and stored at open-circuit potential (OCP) for the same time as the corresponding cycling. The cells were disassembled in the glovebox after cycling/storage, and the electrodes were retrieved. If not stated otherwise, the electrodes are unwashed to preserve any surface species formed. Comparisons were made with an electrode washed in DMC for 5 h and thereafter dried under vacuum for 12 h.

XPS measurements were conducted on a PHI 5500 system, using a monochromatized Al K α anode as X-ray excitation source ($h\nu = 1486.6$ eV). All analyses were carried out under an inert atmosphere, and the electrode samples were carefully transported to the analysis apparatus under an inert atmosphere, using a purpose-built device, to preserve the surface layer formed. The pass energy was 23.5 eV, and the energy step was 0.100 eV. During analysis, the pressure in the chamber was $<10^{-7}$ Pa. Depth information on the elemental composition was obtained by Ar ion beam sputtering. Peak assignments were made by Gaussian–Lorentzian curve fits of the spectra. Calibration of peak position was performed by measuring distances between peaks in the spectra of specific compounds found in the surface layer to avoid charging effects. The peak-fitting procedure was performed with XPSPEAK 4.1 (written by Dr. R. W. M. Kwok, Chinese University of Hong Kong) and MultiPak V6.1A (Physical Electronics, Inc.) software. The final scale determination was performed numerically using the C 1s graphite peak (284.3 eV), from CB in the composite electrode, as reference. Reference measurements were recorded for LiF , Li_2CO_3 , CH_3OLi , LiPF_6 , LiBF_4 , poly(ethylene oxide), and LiMn_2O_4 . It should be noted that XPS is an ex situ technique and placing the electrode samples in a vacuum may alter their surface chemistries.

SEM measurements were made using a LEO 1550 instrument with a field-emission gun (FEG) as electron source.

The DSC samples for cycled and stored cells were taken from cells disassembled in a glovebox and the LiMn_2O_4 cathodes allowed to dry overnight. Small disks were punched out and sealed in standard Al crucibles inside the glovebox. The Al-foil current collector was placed against the Al crucible to ensure minimum contact between the walls and the active sample. The measurements were carried out using a Mettler DSC 30 calorimeter, with the heating rate set at 5 °C/min in the temperature range 0–300 °C. The various components of the half-cells were studied separately and in all possible combinations.

The infrared spectra were measured on a Bruker IFS 66v/S FT-IR spectrometer equipped with a DTGS detector using a diffuse reflectance device (Graseby Specac, mod. Selector). The sample holder was tilted so that the total reflectance was measured. The resolution was 2 cm⁻¹, and 200 scans were accumulated for each spectrum. A crumpled Al foil was used as reference. The data were analyzed by the Kramers–Kronig technique to calculate the absorption spectra reported here.³⁷

The XRD measurements were made using a STOE & CIE GmbH STADI position-sensitive detector (PSD) X-ray powder diffractometer (Cu K α_1 radiation), and the data were collected using a 6° straight-wire detector. The electrodes were measured ex situ, sealed in a protective polymer laminate bag to preserve any atmosphere-sensitive phases.

Results and Discussion

XPS Qualitative Analysis. Figure 1 presents the C 1s spectra for a series of electrode samples. The general shape of the C 1s spectra for the electrode surfaces after cycling or storage at elevated temperature is quite different from that for the ambient temperature samples (Figure 1).

The main C peak is clearly broadened for the electrode cycled at room temperature (RT) compared to that for the

(26) Du Pasquier, A.; Dismas, F.; Bowmer, T.; Gozdz, A. S.; Amatucci, G.; Tarascon, J.-M. *J. Electrochem. Soc.* **1998**, *145*, 472.

(27) Zheng, T.; Gozdz, A. S.; Amatucci, G. G. *J. Electrochem. Soc.* **1999**, *146*, 4014.

(28) Richard, M. N.; Dahn, J. R. *J. Electrochem. Soc.* **1999**, *146*, 2068.

(29) Andersson, A. M.; Edström, K. *J. Electrochem. Soc.*, in press.

(30) Eriksson, T.; Andersson, A. M.; Bishop, A. G.; Gejke, C.; Gustafsson, T.; Thomas, J. O. *J. Electrochem. Soc.*, in press.

(31) Xia, Y.; Zhou, Y.; Yoshio, M. *J. Electrochem. Soc.* **1997**, *144*, 2593.

(32) Blyr, A.; Sigala, C.; Amatucci, G.; Guyomard, D.; Chabre, Y.; Tarascon, J.-M. *J. Electrochem. Soc.* **1998**, *145*, 194.

(33) Gummow, R. J.; de Kock, A.; Thackeray, M. M. *Solid State Ionics* **1994**, *69*, 59.

(34) Amatucci, G. G.; Pereira, N.; Zheng, T.; Tarascon, J.-M. *J. Electrochem. Soc.* **2001**, *148*, A171.

(35) Amatucci, G. G.; Blyr, A.; Sigala, C.; Alfonse, P.; Tarascon, J. M. *Solid State Ionics* **1997**, *104*, 13.

(36) Wang, E. U.S. Pat. 5,733,685, 1998.

(37) Garbun, M. *Optical Physics*; Academic Press: New York, 1965; p 121.

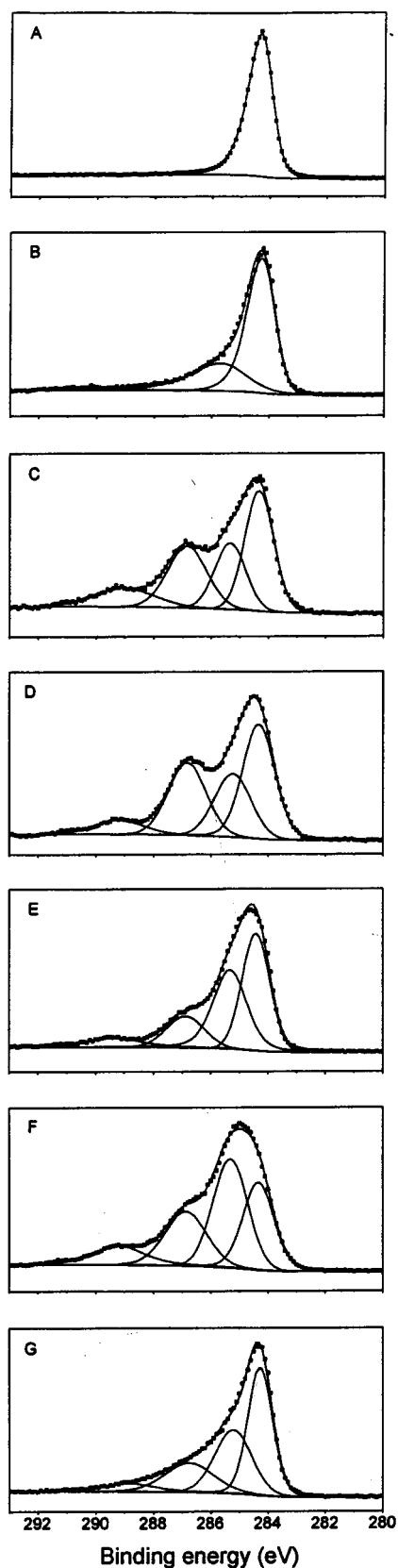
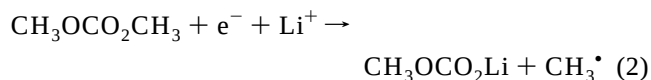
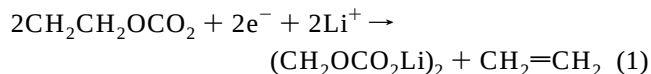


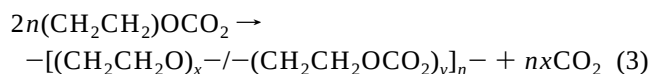
Figure 1. XPS C 1s spectra of LiMn₂O₄ electrode surfaces: (A) pristine electrode; (B) cycled 50 times C/3 at ambient temperature, LiPF₆ electrolyte; (C) cycled 50 times C/3 at 60 °C, LiPF₆ electrolyte; (D) stored 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte; (E) stored 300 h at 60 °C, SOD 0%, LiPF₆ electrolyte; (F) cycled 50 times C/3 at 60 °C, LiBF₄ electrolyte; (G) stored 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte, washed with DMC. Solid lines represent each individual peak and the total fit; the experimental data are represented by squares.

pristine electrode (Figure 1A,B). The C 1s spectra for the cycled RT sample can be reproduced by three peaks at 284.3 eV (graphitic carbon from CB), 285.5 eV, and 290 eV (EC residues). The shoulder at 285.5 eV can be assigned to some polymeric species with an electronegative constituent, e.g., poly(oxyethylene).³⁰ It could also originate from a pure polyethylene compound.^{21,22,24,38}

Additional bands were observed at 286.8 and 288.8–289.4 eV in the elevated temperature stored and cycled samples (Figure 1C–G). These can be assigned to ether and carbonate carbons, respectively, originating from solvent decomposition products. The major reaction product from EC is known to be (CH₂OCO₂Li)₂, while DMC decomposes into CH₃OCO₂Li:²⁰



These products will be referred to hereafter as “semicarboxylates” and have been found to give a C 1s peak shifted toward lower binding energy (BE) compared to the case of Li carbonate.^{24,39} They are therefore a plausible explanation for the observed C 1s peaks. However, rigorous studies have shown semicarboxylates to be unstable at elevated temperatures, transforming into the more stable Li₂CO₃.^{26,28} It is thus unlikely that a stable semicarboxylate surface layer would form at elevated temperatures. A distorted carbonate group with additional bonds to other organic groups could appear at a slightly lower binding energy than Li₂CO₃; the most likely candidate is a polymerized carbonate, which can be formed directly by polymerization of EC initiated by EC oxidation.¹³ Another possibility is the direct polymerization of EC initiated by strong Lewis acids, e.g., PF₅ or BF₃, into a poly(oxyethylene-*alt*-ethylene carbonate):^{40,41}



These types of polymer have been shown to exist in the graphite SEI layer, especially on chemical lithiation and storage at elevated temperature.^{25,29} Reaction 3 explains satisfactorily both the polymer/poly(oxyethylene) band at 285.5 eV and the distorted carbonate bands at 286.8 and 288.8–289.4 eV, although it cannot be established definitively whether one or two polymers are present.

Electrodes treated at 60 °C all exhibit the same bands in different proportions. It is worth noting that considerably more polymeric species are formed using LiBF₄ as electrolyte salt (Figure 1 F) than with LiPF₆, possibly because BF₃ is a stronger initiator of polymerization. After a thorough DMC wash (as described in the Experimental Section), the polymeric feature at 285.3 eV is unaffected, and the polymerized carbonate peaks at 286.8 and 288.8–289.4 eV are still clearly present (Figure 1G).

The O 1s measurements on the reference LiMn₂O₄ electrode show the spinel oxygen at 529.5 eV and a broad

(38) Aurbach, D.; Markovsky, B.; Weissman, I.; Levi, E.; Ein-Eli, Y. *Electrochim. Acta* **1999**, *45*, 67.

(39) Shiraishi, S.; Kanamura, K.; Takehara, Z. *J. Electrochem. Soc.* **1999**, *146*, 1633.

(40) Vogdanis, L.; Martens, B.; Uchtmann, H.; Hensel, F.; Heitz, W. *Makromol. Chem.* **1990**, *191*, 465.

(41) Vogdanis, L.; Heitz, W. *Makromol. Chem., Rapid Commun.* **1986**, *7*, 543.

feature at higher binding energies which fits to two Lorentzian peaks (Figure 2A). This shoulder originates from surface hydroxyl and adsorbed water, although all electrodes were dried in a vacuum furnace at 120 °C for >4 h prior to use to minimize water adsorption. Two broad bands at 532.7–533.1 and 534.2–534.6 eV dominate the spectra for cycled and stored electrodes (Figure 2). These are equal in area at RT, where the high BE peak can be associated mainly with EC residues and the low BE peak with polymeric (C–O–C) oxygen³⁹ and phosphorus pentoxide.³⁰

The low BE O 1s peak is dominant for electrodes cycled or stored at 60 °C, although it decreases somewhat in intensity upon thorough DMC washing (Figure 2C–G). This can be associated mainly with the large amount of P in the surface film (Figure 5). The polycarbonate/poly-(oxyethylene) peaks appear at 532–533 and 534–534.5 eV, and the low BE ether band is partly masked by the phosphorus oxide peak. The LiMn_2O_4 oxygen peak at 529.5 eV is only barely distinguishable for all samples cycled and stored at above-ambient temperature (Figure 2C–G); its signal is suppressed by that from electrolyte decomposition products covering the surface.

The P 2p spectrum for a sample cycled at ambient temperature has a band at 138 eV (from a Li_xPF_y compound) and a shoulder at 135.4 eV, assigned to a phosphorus oxide compound, e.g., P_2O_5 (Figure 3B).^{42,43} After cycling or storage at 60 °C, the 138 eV peak disappears, and a new peak appears at 136.5 eV (Figure 3C–F). The 136.5 eV peak can be related to the increase in F-containing compounds (686–688 eV); this reflects the increase in $\text{Li}_x\text{PO}_y\text{F}_z$ -type rather than Li_xPF_y -type compounds as found for ambient temperature samples.

There is a strong contribution from the 135.1–135.3 eV peak for electrodes cycled or stored at state-of-discharge (SOD) 100% (i.e., LiMn_2O_4) at 60 °C (Figure 3B,C,E); this is less prominent for the SOD 0% sample (i.e., $\lambda\text{-MnO}_2$) stored at 60 °C (Figure 3D). This peak coincides well with P_2O_5 , together with the broad oxygen band at 533 eV.⁴³ The broadness of the oxygen band may result from the presence of two adjacent oxygen bands associated with P_2O_5 : from nonbridging (P–O bonds) and bridging oxygens (P–O–P bonds). The magnitude of the P 2p P_2O_5 peak decreases somewhat after a thorough DMC wash (Figure 3E), coinciding with a decrease in the O 1s P_2O_5 peak (Figure 2G).

When LiBF_4 was used, a B 1s band was observed with a peak at 196 eV from $\text{Li}_x\text{BO}_y\text{F}_z$ compounds and a shoulder at 194 eV from boron oxides, e.g., B_2O_3 (Figure 3F).⁴²

XPS spectra for electrode surfaces from four different pretreatment conditions are shown in Figure 4; similar species are formed in only slightly different amounts. The F 1s spectra contain a strong, broad peak at 686–688 eV, possibly comprising more than one peak, showing that there are a number of $\text{Li}_x\text{PO}_y\text{F}_z$ species on the surface.¹⁸ There is also a shoulder at 685.5 eV from LiF .^{18,21,22} This becomes the dominant feature on sputtering. There is almost no detectable Mn on the surface of the electrodes, due to the thickness of the surface film formed. Sputtering for 60 s removes enough of this film for Mn to be observed. However, since sputtering also removes oxygen from the spinel structure, it is not possible to determine the oxidation state of the spinel. Lithium has a peak at 56–57 eV.

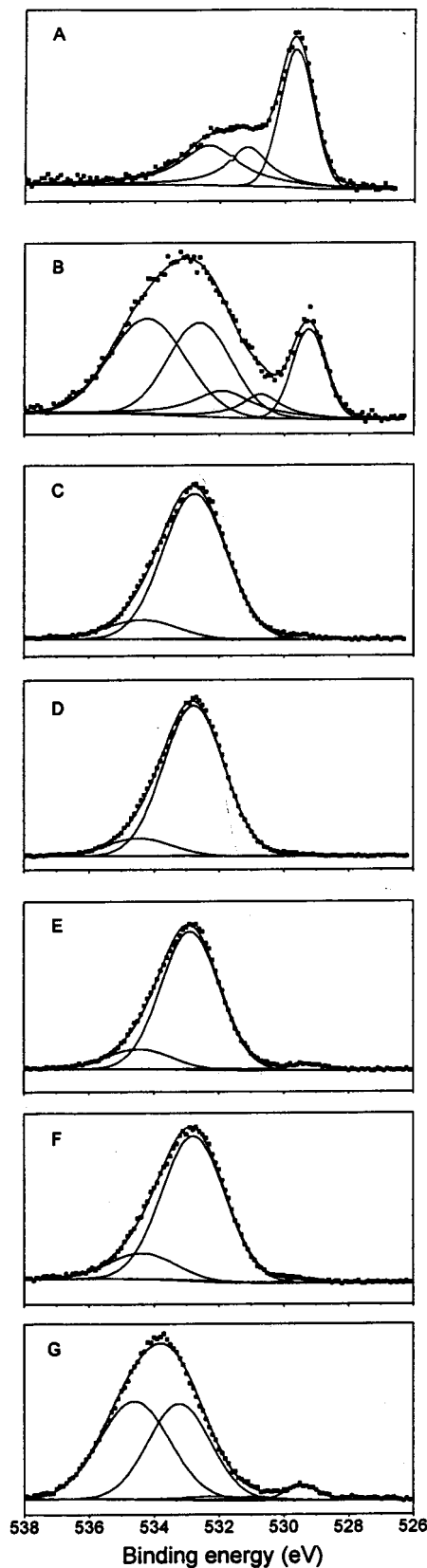


Figure 2. XPS O 1s spectra of LiMn_2O_4 electrode surfaces: (A) pristine electrode; (B) cycled 50 times C/3 at ambient temperature, LiPF_6 electrolyte; (C) cycled 50 times C/3 at 60 °C, LiPF_6 electrolyte; (D) stored 300 h at 60 °C, SOD 100%, LiPF_6 electrolyte; (E) stored 300 h at 60 °C, SOD 0%, LiPF_6 electrolyte; (F) cycled 50 times C/3 at 60 °C, LiBF_4 electrolyte; (G) stored 300 h at 60 °C, SOD 100%, LiPF_6 electrolyte, washed with DMC. Solid lines represent each individual peak and the total fit; the experimental data are represented by squares.

(42) Wagner, C. D.; Riggs, W. M.; Davis, L. E.; Moulder, J. F.; Muilenberg, G. E. *Handbook of X-ray Photoelectron Spectroscopy*; Perkin-Elmer Physical Electronics Division: Eden Prairie, MN, 1979.

(43) Khattak, G. D.; Salim, M. A.; Wenger, L. E.; Gilani, A. H. J. *Non-Cryst. Solids* **2000**, 262, 66.

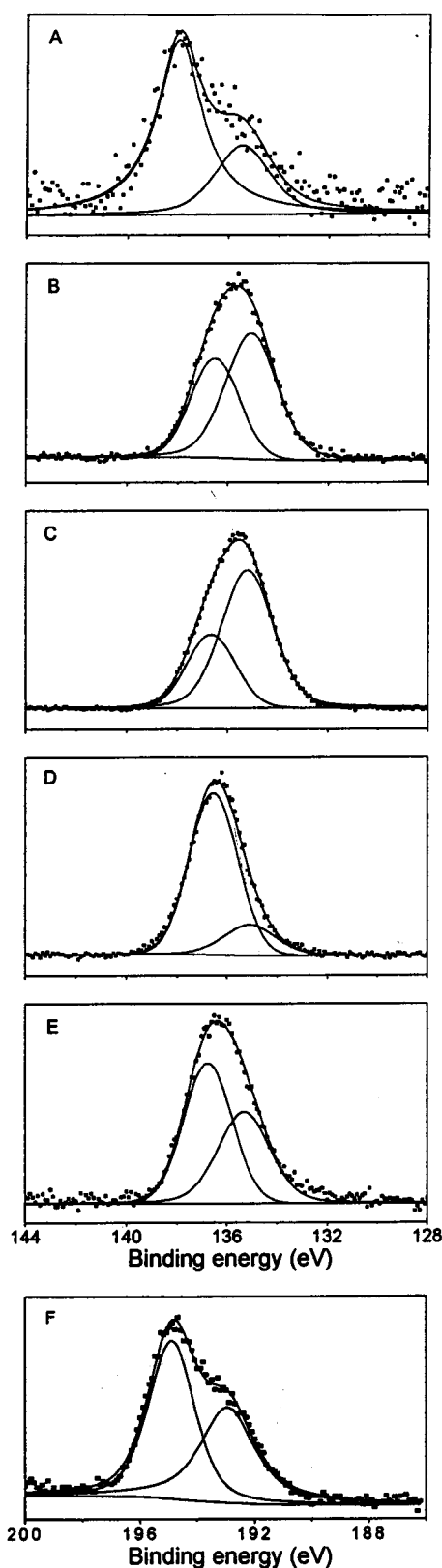


Figure 3. XPS P 2p/B 1s spectra of LiMn₂O₄ electrode surfaces: (A) cycled 50 times C/3 at ambient temperature, LiPF₆ electrolyte; (B) cycled 50 times C/3 at 60 °C, LiPF₆ electrolyte; (C) stored 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte; (D) stored 300 h at 60 °C, SOD 0%, LiPF₆ electrolyte; (E) stored 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte, washed with DMC; (F) cycled 50 times C/3 at 60 °C, LiBF₄ electrolyte. Solid lines represent each individual peak and the total fit; the experimental data are represented by squares.

XPS Depth Profile and Quantitative Analysis. A series of sputtering experiments for a cycled and a stored electrode are shown in Figure 5, where 0, 60, 120, 300, 600, and 1500 s sputter times have been analyzed. The atomic fraction of a specific element (C_x) is given by the general formula

$$C_x = \frac{n_x}{\sum_i n_i} = \frac{I_x/S_x}{\sum_i I_i/S_i} \quad (4)$$

where S is the atomic sensitivity factor, a factor dependent on the element and the instrumental setup; n is the number of atoms of the element per cm³ of the sample; and I is the number of photoelectrons per second in a specific spectral peak.⁴²

Initially, both electrodes show a small amount of C and Mn on the surface and a high concentration of O, P, F, and Li. The C and Mn concentrations then increase continuously as the surface layer is removed. Li increases slightly in both samples after 60 s sputtering but then decreases, together with the other surface-related elements O, P, and F. The amounts of Li and O level out as Mn increases, and more LiMn₂O₄ is brought to the surface. P decreases to practically zero, while F is surprisingly difficult to remove from the electrode surface and levels out at around 10–15 at. %. This shows that F probably redeposits on the electrode surface during sputtering.

There is a striking resemblance in the surface chemistry of an electrode cycled 50 times at C/3 in 60 °C and one stored for 300 h in 60 °C. The electrochemical contribution would thus seem to be minimal in the formation of the surface film at elevated temperatures.

The amounts of each element on the unsputtered surfaces are summarized in Table 1. One should bear in mind that, although photoionization occurs to a depth of several micrometers, only electrons from within tens of angstroms below the surface can escape without inelastic scattering energy loss. XPS thus monitors a region only a few nanometers below the electrode surface. Practically all the species on the electrodes originate from the surface layer, as can be seen from a comparison with the measurement on a pristine LiMn₂O₄ electrode (Figures 1–5). However, a contribution from the nanocrystalline, turbostratic graphite regions in CB is present in all samples. The concentration values given within parentheses are after the subtraction of this graphite peak and thus correspond only to the amount of carbon associated with surface species (Table 1).

It is evident that large amounts of O, P, F, and Li collect at the surface during cycling or storage at SOD 100% (LiMn₂O₄), 60 °C, compared to other samples. Some of the P–O products can be removed through thorough washing with DMC, and a surface containing more of the polymeric C species is revealed. After storage at SOD 0% (λ-MnO₂), the P–O products are less pronounced, and instead, more polymeric products are observed. The same is true of the electrode cycled in LiBF₄ electrolyte, which has a similar elemental composition to the stored λ-MnO₂ electrode (with B substituted for P compounds). The lower amounts of B-containing compounds compared to phosphorus oxides can be explained by the greater instability of the LiPF₆ salt, making it more sensitive to impurities in the electrolyte.

The Li concentration increases after washing (or sputtering), indicating that a Li-rich layer is present under the outer P–O layer; LiF is the most likely product. It should be mentioned that a surplus of F is always found

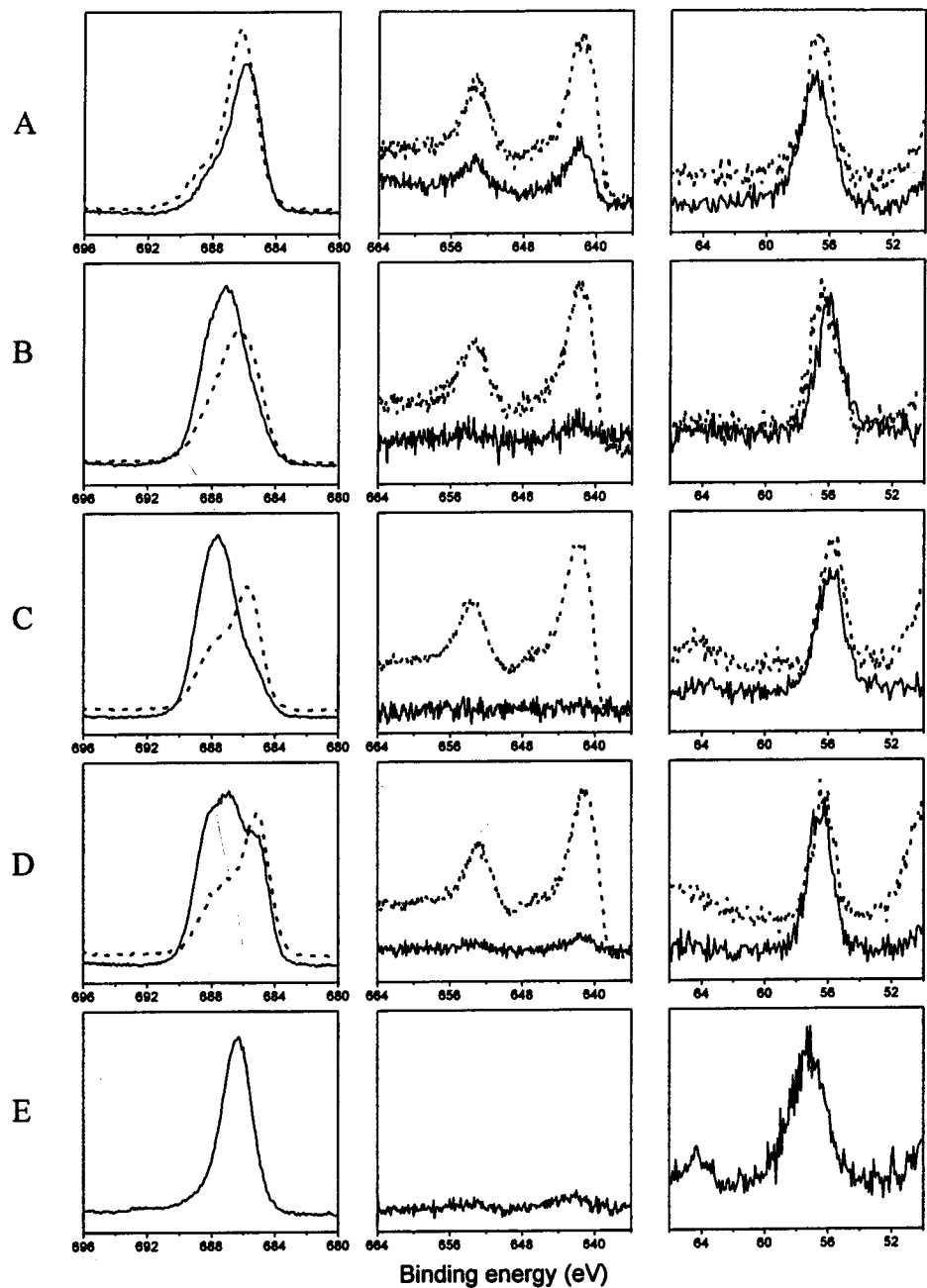


Figure 4. XPS F 1s, Mn 2p, and Li 1s spectra of LiMn₂O₄ electrode surfaces: (A) cycled 50 times C/3 at ambient temperature, LiPF₆ electrolyte; (B) cycled 50 times C/3 at 60 °C, LiPF₆ electrolyte; (C) stored 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte; (D) stored 300 h at 60 °C, SOD 0%, LiPF₆ electrolyte; (E) cycled 50 times C/3 at 60 °C, LiBF₄ electrolyte. Full lines unsputtered; dashed lines 60 s Ar⁺ sputtering.

Table 1. Atomic Percentage (at. %) of Elements on the Surface of the Electrode Samples, EC/DMC 2:1 1 M LiPF₆ or LiBF₄^a

electrode sample	element						
	C 1s	O 1s	P 2p	B 1s	F 1s	Li 1s	Mn 2p
LiMn ₂ O ₄ , 50 cycles, C/3, 60 °C, LiPF ₆	29.1 (17.8)	22.6	7.6		30.5	9.9	0.3
LiMn ₂ O ₄ , 300 h storage, 60 °C, LiPF ₆	25.6 (15.4)	24.1	9.3		31.5	9.3	0.2
λ-MnO ₂ , 300 h storage, 60 °C, LiPF ₆	46.3 (30.6)	18.0	5.1		23.8	6.5	0.3
LiMn ₂ O ₄ , 300 h storage, 60 °C, LiPF ₆ , washed	42.9 (24.6)	11.8	3.1		22.8	19.1	0.3
LiMn ₂ O ₄ , 50 cycles, C/3, 60 °C, LiBF ₄	52.0 (37.3)	19.9		5.3	16.4	6.2	0.2
LiMn ₂ O ₄ , 50 cycles, C/3, RT, LiPF ₆	55.3 (18.2)	5.7	1.0		20.9	16.2	0.9
LiMn ₂ O ₄ pristine	90.3	6.4				0.6	2.7

^a Carbon content in parentheses is without the contribution of the graphite peak, i.e., only surface film carbon.

on the electrodes, making it impossible to calculate exact amounts of the surface species. There is no other possible explanation than the formation of a fluorinated carbon compound, and this would give C 1s peaks at BE > 290

eV, which is not observed in any samples. We have no explanation for the discrepancy at this time.

The Mn signal is barely detectable in the 60 °C samples, and washing does not seem to improve the Mn signal; the

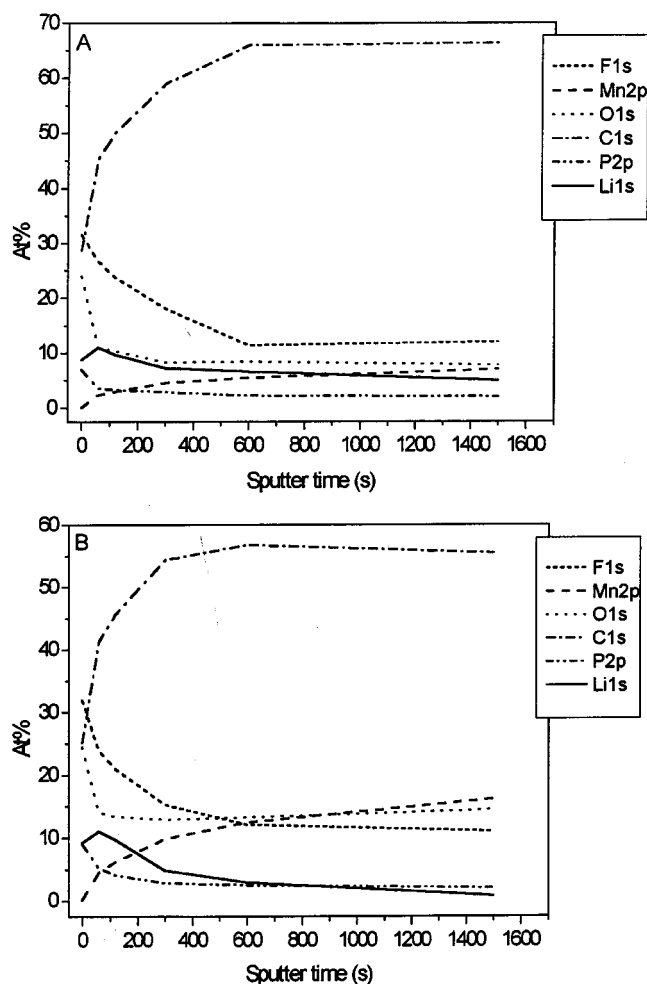


Figure 5. XPS determined elemental depth profiles of LiMn₂O₄ electrodes: (A) cycled 50 times C/3 at 60 °C, LiPF₆ electrolyte; (B) stored 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte.

polymeric/polycarbonate species clearly coat the surface too efficiently. We can thus conclude from the XPS studies that surface film thickness increases with temperature. While the graphite anode SEI layer would seem to break down at slightly elevated temperatures, and have to be re-formed in the subsequent reduction cycle,²⁹ the cathodic surface layer increases in thickness. The anodic SEI layer, composed of a thin inner layer of insoluble Li salts and an outer layer comprising a more porous blend of organic material, passivates the reactive graphite surface at some thickness. Further electrolyte reduction processes when the electrode is subsequently polarized to low potentials are hence suppressed. The surface of the anodic electrode is then electronically insulated but allows Li ions to migrate through it. In the case of the spinel cathode surface film, a different mechanism must be responsible for surface film growth. The thickness of the cathode surface film increases with storage time, number of cycles, and temperature. This implies that the inner part of the cathode surface film must lack the dense Li salt component found on the anode side, thus allowing solvent to penetrate to the Helmholtz layer of the active cathode surface, where further polymerization reactions can take place.

DSC Measurements. DSC peaks at 20 °C correspond to EC/DMC melting and above 200 °C to salt and solvent decomposition. DSC peaks for all electrode samples which have been in contact with the LiPF₆ electrolyte also show one exothermic peak between 140 and 160 °C and another between 260 and 280 °C (Figure 6B). These cannot be

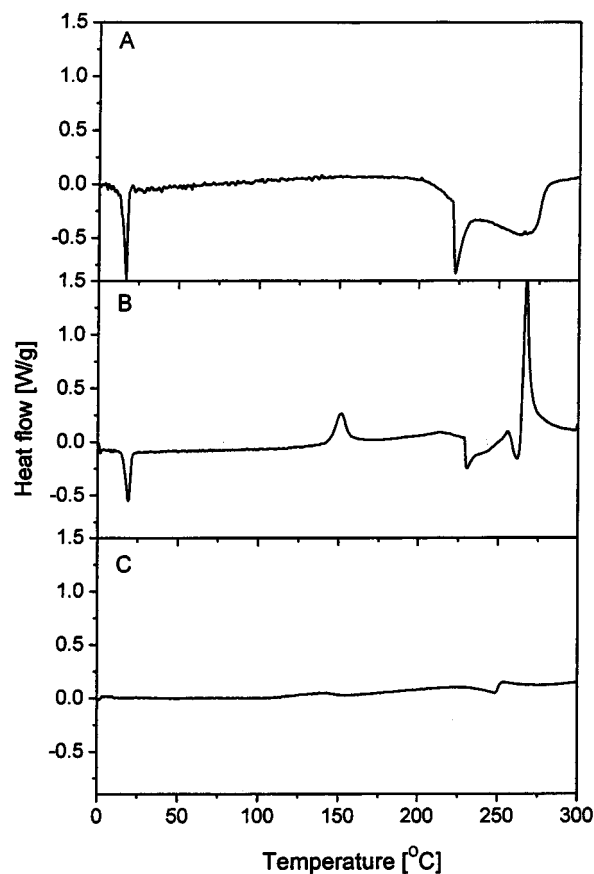


Figure 6. DSC measurements of composite electrodes: (A) CB/EPDM 90/10 w/w electrode polarized to 4.5 V, LiPF₆ electrolyte; (B) LiMn₂O₄/CB/EPDM 80/15/5 w/w electrode, cycled in LiPF₆ electrolyte; (C) LiMn₂O₄/CB/EPDM 80/15/5 w/w electrode, cycled in LiBF₄ electrolyte.

related to the electrolyte alone or to any structural changes in pure LiMn₂O₄, nor can they be contributions from carbon black or binder.⁴⁴ Moreover, the reactions associated with these peaks are not reversible; no corresponding peaks are observed on cooling. The only options that remain are gas evolution and some irreversible reaction(s). Gas evolution usually causes a shift in DSC baseline, but the baseline is flat here. Moreover, the peak shape at ca. 150 °C is quite broad and symmetric, indicating some sort of reaction. The enthalpy and peak position are very similar to the first reaction peak in a series of reactions previously observed for carbon anodes.²⁶ The carbon electrode SEI peak is superimposed on a peak corresponding to a reaction between Li_xC₆ and electrolyte; the reaction constitutes a large contribution to the observed enthalpy (ΔH_1 in ref 26). The decomposition of the SEI layer on carbon electrodes is generally related to reactions involving semicarbonates, e.g., R-O-C-O₂-Li, to form a more stable lithium-carbonate surface film.²⁸ However, no such semicarbonates could be found as surface products on LiMn₂O₄ electrodes after cycling and storage in an EC/DMC electrolyte at ambient temperature; it would thus seem unlikely that similar reactions occur on the cathode as on the anode.

The position and integrated intensity (ΔH) do not change markedly for the 150 °C peak between a series of cycled or stored electrodes, and the enthalpy for the reaction is quite low (Figure 7). The 270 °C peak is also sharp and narrow and overlaps with the endothermic electrolyte

(44) Fransson, L.; Eriksson, T.; Edström, K.; Gustafsson, T.; Thomas, J. O. J. *Power Sources*, in press.

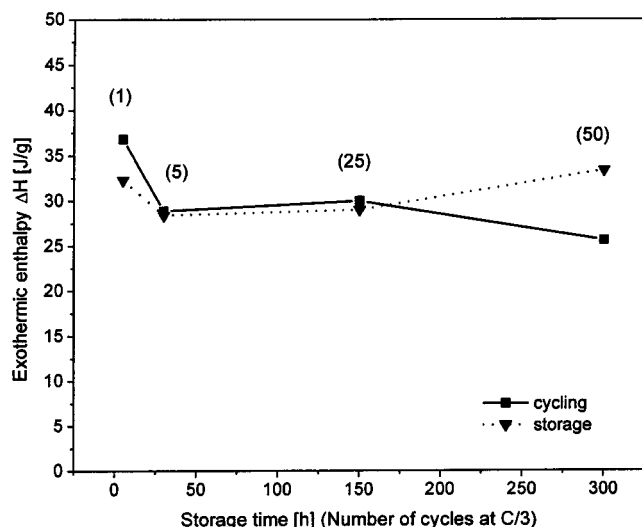


Figure 7. Enthalpy for the 150 °C peak, calculated from DSC traces on a series of stored and cycled LiMn_2O_4 electrodes.

decomposition. The probable explanation for both exothermic peaks in the 100–300 °C region is a reaction occurring between Li in the active cathode material and the electrolyte.⁴⁵

Stored electrodes were studied at different state-of-discharge (SOD) (Figure 8); a steady increase in reaction onset temperature is observed as a function of SOD (Figure 9). The enthalpy shows different behavior, with maxima occurring between the ordered single-phase structures at SOD 0%, 50%, and 100% (Figure 9). Electrodes stored and cycled at 60 °C show an even higher reaction onset temperature (Figure 8, bottom). This can thus be interpreted in terms of the formation of some lithium-rich surface structure on the electrodes.

No onset temperature was found that could be related to surface layer formation. Higher temperatures thus seem to result rather in a general increase in reaction kinetics than in promoting new reactions in addition to those taking place at ambient temperature.

SEM Measurements. SEM measurements show no significant differences between samples cycled 50 times with LiPF_6 or LiBF_4 salts at 60 °C or ambient temperature. The electrolyte residues cannot be distinguished, nor is major LiF formation detected, as is normally the case for precycled and stored graphite anodes.

Reflectance FT-IR Measurements. Notable differences are seen between a reference electrode (electrolyte applied to a LiMn_2O_4 surface) and an electrode stored 300 h at 60 °C (Figure 10, Table 2). A clear divergence from the reference-sample spectrum is the band in the $\nu(\text{C}-\text{O})$ region at 1167 cm^{-1} ; this is possibly from a polycarbonate or poly(oxyethylene) species.^{25,46,47} Another clear difference is the broad composite band with centers at 1270 – 1314 cm^{-1} , followed by a broad, less intense band at 1630 – 1650 cm^{-1} . These bands can also be associated with polycarbonate or Li-alkyl carbonates; $\nu(\text{C}=\text{O}_s)$ gives bands around 1250 – 1350 cm^{-1} and $\nu(\text{C}=\text{O}_{as})$ bands at 1600 – 1700 cm^{-1} .^{13,25,46} The intensity of the band at 716 cm^{-1} also increases, and a new band appears at 928 cm^{-1} ; the band at 981 cm^{-1} shifts to 975 cm^{-1} , and another new band appears at 1393 cm^{-1} ; finally, the intensity of the

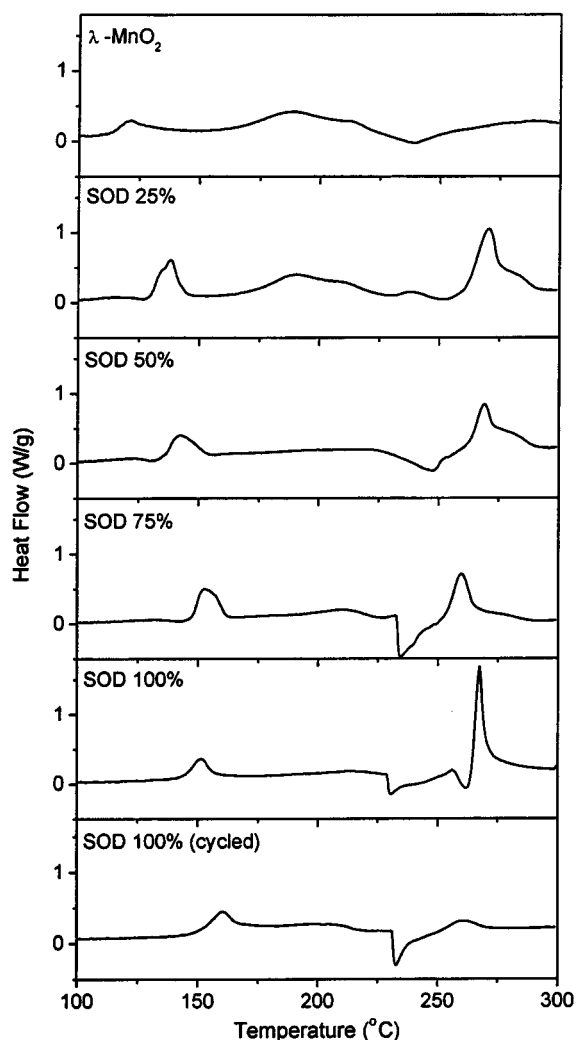


Figure 8. DSC measurements on a series of electrodes from cells stored at different degrees of SOD. Bottom: DSC trace for an electrode cycled 50 times C/3 at 60 °C, LiPF_6 electrolyte.

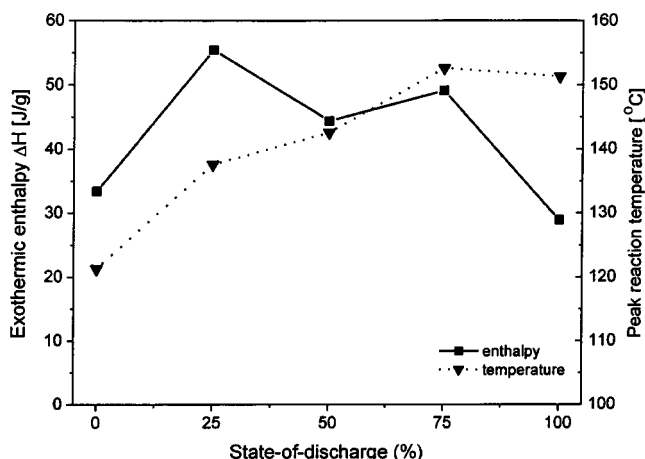


Figure 9. Enthalpy and main-peak temperature, calculated from Figure 8.

shoulder at 1730 cm^{-1} also increases. The 1393 cm^{-1} band and part of the broad 1270 – 1314 cm^{-1} band can be associated with the weak $\delta(\text{CH}_3, \text{CH}_2)$ vibrations known to occur at 1250 – 1450 cm^{-1} in polycarbonates, Li-alkyl carbonates, and poly(oxyethylene).^{25,46,47}

All the new bands could not be explained by the presence of polycarbonate/poly(oxyethylene). A strong $\nu(\text{P}-\text{F})$ band from Li_xPF_y occurs at 842 cm^{-1} ; this band becomes much

(45) Zhang, Z.; Fouchard, D.; Rea, J. R. *J. Power Sources* **1998**, *70*, 16.

(46) Soga, K.; Hosada, S.; Tazuke, Y.; Ikeda, S. *J. Polym. Sci., Polym. Lett.* **1976**, *14*, 161.

(47) Yoshihara, T.; Todokoro, H.; Murahashi, S. *J. Phys. Chem.* **1964**, *41*, 2902.

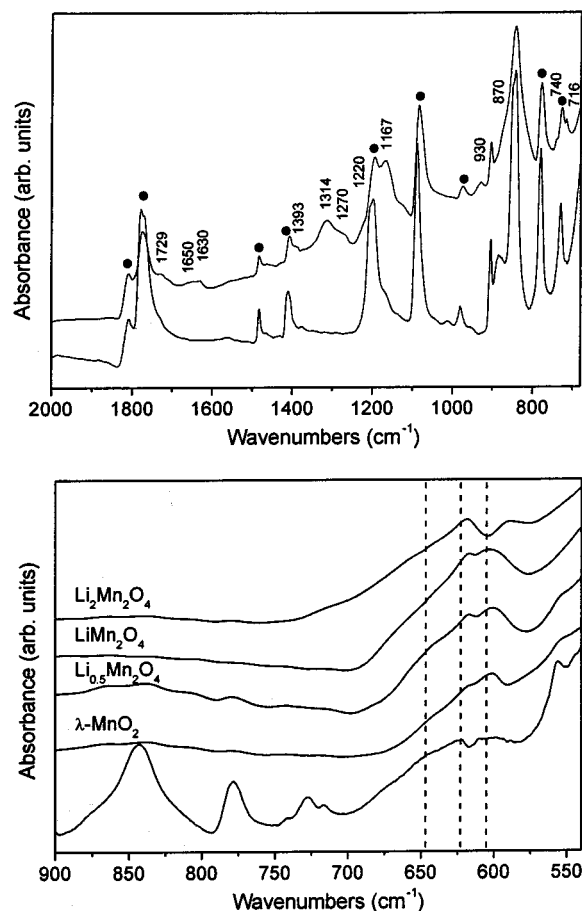


Figure 10. Total reflectance FT-IR spectra for a LiMn₂O₄ electrode stored for 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte: (A) 680–2000 cm⁻¹. Upper spectrum from the sample. Lower spectrum from electrolyte applied on a pristine electrode surface and measured. EC bands indicated with dots. (B) 540–900 cm⁻¹. Comparison between [Mn–O₆] octahedral bands from a stored electrode at 60 °C and reference electrodes.

Table 2. Assignment of Peaks in Infrared Spectra of LiMn₂O₄ Samples^a

frequency (cm ⁻¹)	proposed assignment ^b
600–605	[Mn–O ₆] LiMn ₂ O ₄
615–620	[Mn–O ₆] LiMn ₂ O ₄
620–630	[Mn–O ₆] Li _{1+x} Mn _{2-y} O ₄
645	[Mn–O ₆] LiMn ₂ O ₄
740	P–O–P band P _x O _y
842	ν(P–F) Li _x PF _y
870 sh	ν(P–F), ν(P–O) Li _x PO _y F _z
930	ν(P–F), ν(P–O) Li _x PO _y F _z or P _x O _y
1167	ν(C–O–C)
1175	EC, Li _x PO _y F _z or P _x O _y
1270	(CH ₂)
1285	Li _x PO _y F _z or P _x O _y
1314	ν(C=O) _s
1287	(CH ₂)
1332	(CH ₂)
1393	(CH ₃ ,CH ₂) or P _x O _y
1630	Li _x PF _y or Li _x PO _y F _z
1650	ν(C=O) _{as}
1729	ν(C=O)

^a Based on refs 13, 19, 22, 24, 38, and 46–50. ^b ν, stretching; sh, shoulder.

broader for the stored electrode, and a shoulder forms at around 870 cm⁻¹. The P–F environment has clearly changed; a possible overlap with the δ(OCO₂⁻) vibration from a carbonate compound has also developed. However, the change in the band at 731 cm⁻¹ to a broad band with

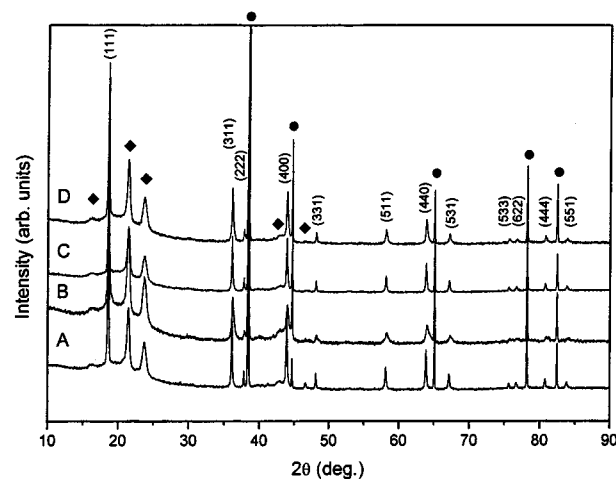


Figure 11. Ex situ XRD diffractograms for LiMn₂O₄ electrodes: (A) LiMn₂O₄ reference electrode; (B) cycled 50 times C/3 at 60 °C, LiPF₆ electrolyte; (C) stored 300 h at 60 °C, SOD 100%, LiPF₆ electrolyte; (D) stored 300 h at 60 °C, SOD 0%, LiPF₆ electrolyte. Solid circles indicate Al current-collector peaks, and solid diamonds indicate peaks from the protective polymer bag.

peaks at 716, 727, and 740 cm⁻¹ can have its origin in a P–O vibration in a phosphorus oxide compound, e.g., Li_xPO_yF_z. Bands arising at 930, 1175, 1270, 1393, and 1630 cm⁻¹ can also be assigned to Li_xPO_yF_z or other phosphorus oxide compounds.^{25,48–50} These findings support our results from the XPS measurements.

The above discussion around solvent polymerization has been directed mainly toward reactions involving EC since EC is more likely to be oxidized at the positive electrode. Its dielectric constant is much higher than that for DMC (95.3 compared to 3.12); PF₆⁻ anions will thus be solvated preferentially by EC. When a positive potential is applied, the concentration of anions—and with it the concentration of EC—will increase in the double layer at the positive electrode.⁵¹ The strong electrostatic field close to the cathode surface will serve to align the polar EC molecules and facilitate oxidation. Because of its higher polarity, EC will also be the preferred target of electrophilic and nucleophilic attack by contaminants in the electrolyte solution.⁵¹

A shift was observed in the Mn–O₆ octahedral band toward higher wavenumbers (Figure 10B). This was also reported for the ambient temperature samples stored and cycled for a prolonged time. It is evident that there are similarities between the bands from LiMn₂O₄, Li₂Mn₂O₄, and the electrode sample, but the composition would appear to be somewhat different.

XRD Measurements. Ex situ XRD studies of the electrodes show that no new bulk phases appear in samples cycled or stored at 60 °C (Figure 11). The electrode stored at SOD 0% self-discharges continuously and after 300 h is completely lithiated to LiMn₂O₄ (Figure 11D). The potential curve for the self-discharge corresponds to the normal LiMn₂O₄–λ-MnO₂ galvanostatic cycling curve. XRD phase analysis gives an average composition of the bulk material in the electrode. To obtain a detailed surface phase description, one would have to analyze the specific XPS Mn (Mn 2p, Mn 3p, and Mn 3s) and O peaks associated

(48) Meyer, K. J. *Non-Cryst. Solids* **1997**, 209, 227.

(49) Salim, M. A.; Khattak, G. D.; Sakhawat Hussain, M. J. *Non-Cryst. Solids* **1995**, 185, 101.

(50) Ostrovskii, D.; Ronci, F.; Scrosati, B.; Jacobsson, P. J. *Power Sources* **2001**, 94, 183.

(51) Joho, F.; Novák, Petr. *Electrochim. Acta* **2000**, 45, 3589.

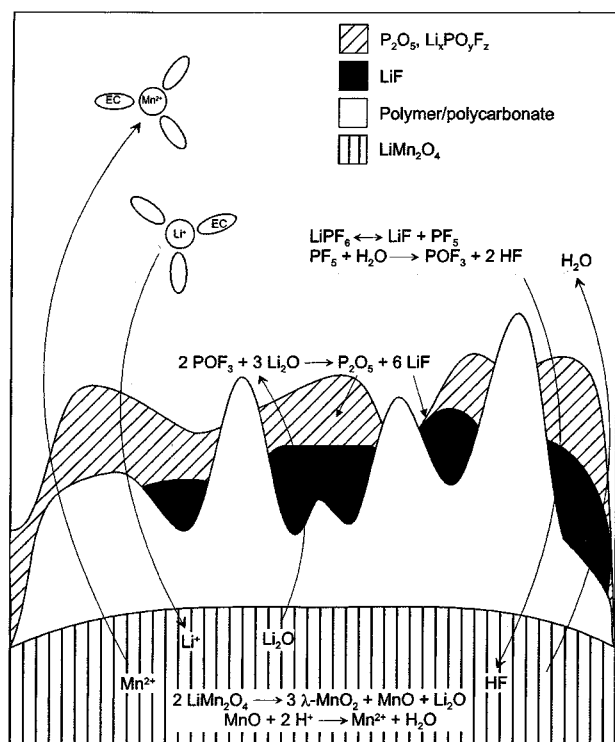
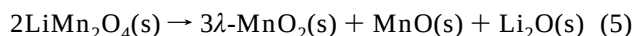


Figure 12. A schematic model of the surface-layer components formed on a LiMn_2O_4 electrode cycled or stored at SOD 100%, 60 °C, and EC/DMC LiPF_6 electrolyte.

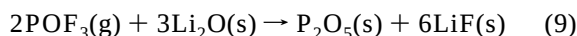
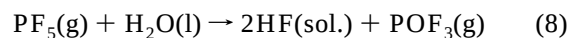
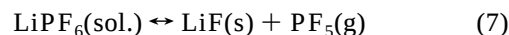
with Li–Mn–O compounds. This was not possible for any electrodes cycled or stored above ambient temperature, due to surface-film coverage preventing a detectable signal emerging from the underlying Li–Mn–O.

Surface Film Model and Mechanisms. On the basis of the above discussion, we propose a model for the formation of different species at the cathode electrode/electrolyte interface (Figure 12). During cycling or storage at SOD 100%, a layer of polymer/polycarbonate is formed closest to the active electrode–particle surface, followed by a layer of LiF and an outermost region of precipitated $\text{Li}_x\text{PO}_y\text{F}_z$ and phosphorus oxides ($\text{Li}_x\text{BO}_y\text{F}_z$ and boron oxides for LiBF_4) (Table 1, Figures 1–4). At SOD 100%, where Mn has a mixed valence state of $\text{Mn}^{3+}/\text{Mn}^{4+}$, a disproportionation of Mn^{3+} occurs with a loss of Mn from the material. This process is described by Hunter's reaction:⁵²



Mn^{2+} goes into solution by forming complexes with the solvent molecules in the electrolyte. Since the LiMn_2O_4 powder used in this study has a composition close to stoichiometric spinel ($\text{Li}_{1.02}\text{Mn}_{1.98}\text{O}_4$ by ICP),³⁰ Hunter's reaction should be followed and not the $\text{Li}^+ \rightarrow \text{H}^+$ exchange reaction.^{14,53,54}

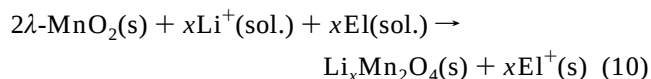
Li_2O from eq 5 can react directly with phosphorus compounds, formed from the decomposition of LiPF_6 ,²⁰ to form P_2O_5 according to



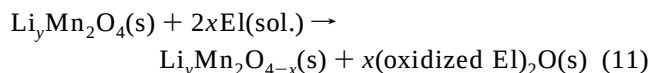
The formation of phosphorus oxide is thus driven by the disproportionation of Mn^{3+} . This would explain the surplus of phosphorus oxides on the SOD 100% samples compared to the SOD 0% samples. The phosphorus oxide in eq 9, as well as that referred to in the XPS discussion, has been written as pure P_2O_5 . While XPS clearly shows a phosphorus(V) oxide, the shifts in the IR bands indicate a somewhat modified phosphorus oxide structure. Pure phosphorus(V) oxide is very reactive; small amounts of Li and F contaminants are thus to be expected and would give rise to IR shifts from stoichiometric P_2O_5 .

Equations 5, 6 and 7, 8 constitute a cycle, whereby H_2O is consumed in eq 8 and re-formed in eq 6. In the LiBF_4 case, a smaller amount of boron oxide is formed through the greater stability of LiBF_4 .

XPS shows there is a stronger driving force for the formation of the polymer/polycarbonate compounds at SOD 0% than at SOD 100%. The initial potential is higher during storage at SOD 0%, and thus more solvent oxidation is expected. Electrodes stored at SOD 0% are also seen to self-discharge back to SOD 100% spontaneously during the storage at 60 °C for 300 h. Electrolyte oxidation has been coupled to Li insertion, and the driving force for electrolyte (El) oxidation should be stronger than when only small changes occur in the lithium content:⁵⁵

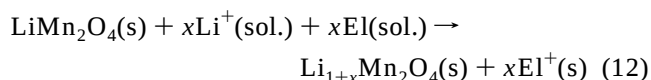


This oxidation may well occur in the form of solvent polymerization. Another possibility is the oxidation of electrolyte coupled to a loss of oxygen from the spinel structure at highly delithiated spinel surfaces:⁵⁶

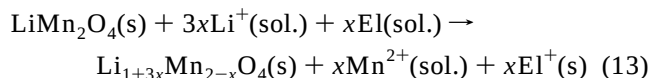


This reaction was proposed to occur in parallel with eq 10.⁵⁶

The changes in the Mn–O₆ octahedra, observed by FT-IR for SOD 100% samples, reflect a shift toward a lithium-rich surface, e.g., the tetragonal $\text{Li}_2\text{Mn}_2\text{O}_4$ phase as suggested previously.^{57,58}



Reaction 12 does not consider the dissolution of Mn^{2+} from the structure, which requires the formulation of $\text{Li}_{1+x}\text{Mn}_{2-y}\text{O}_4$, where $y < 2x$.^{14,30}



(52) Hunter, J. C. *J. Solid State Chem.* **1981**, 39, 142.

(53) Amundsen, B.; Aitchinson, P. B.; Burns, G. R.; Jones, D. J.; Rozière, J. *Chem. Mater.* **1995**, 7, 2151.

(54) Endres, P.; Ott, A.; Kemmler-Sack, S.; Mayer, H. A.; Praas, H.-W.; Brandt, K. *J. Power Sources* **1997**, 69, 145.

(55) Guyomard, D.; Tarascon, J. M. *Solid State Ionics* **1994**, 69, 222.

(56) Gao, Y.; Dahn, J. R. *Solid State Ionics* **1996**, 84, 33.

(57) Thackeray, M. M.; Yang, S.; Kahaian, A. J.; Kepler, K. D.; Skinner, E.; Vaughey, J. T.; Hackney, S. A. *Electrochem. Solid State Lett.* **1998**, 1, 7.

(58) Cho, J.; Thackeray, M. M. *J. Electrochem. Soc.* **1999**, 146, 3577.

Evidence for these reactions cannot be detected by XRD methods, which would suggest that they only occur in the outermost layers of the LiMn₂O₄ particles. The resulting level of Li insertion and the associated amounts of solvent-derived products at SOD 100% must be much lower than those associated with self-discharge of the SOD 0% samples, in good agreement with quantitative XPS analysis.

Conclusions

A model has been presented for the formation of different species on LiMn₂O₄ electrodes after storage and cycling in an EC/DMC-based electrolyte with LiPF₆ or LiBF₄ salt. The formation of poly(oxyethylene-*alt*-ethylene carbonate), LiF, phosphorus oxide, and Li_xPO_yF_z (boron oxide and Li_xBO_yF_z for LiBF₄) has also been established. The polymer precipitates closest to the electrode surface, thereafter LiF, with the Li_xPO_yF_z and phosphorus oxide closest to the electrolyte. From DSC and FT-IR, a Li-rich surface of the Li-Mn-O compound was found.

A number of factors thus affect the growth of the cathode surface layer in a Li battery: temperature, storage potential, and native surface films (not present on the electrodes in this work). Electrochemical cycling would *not* appear to influence the species found in the surface film.

The lack of evidence for an onset temperature associated with the formation of the surface layer suggests that elevated temperatures simply enhance the kinetics of the solvent reactions, rather than induce new reactions compared to those occurring at ambient temperature.

The thickness of the cathode surface film increases with storage time, number of cycles, and temperature. Since the innermost part of the surface film is composed of organic material, it can allow further solvent to approach the spinel surface and continue the breakdown of the electrolyte.

To overcome the solvent oxidation on the cathode and prevent an increase in total cell impedance, it is crucial to obtain a stable inorganic surface layer on the cathodes—preferably prior to cell assembly and contact with electrolyte.

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